

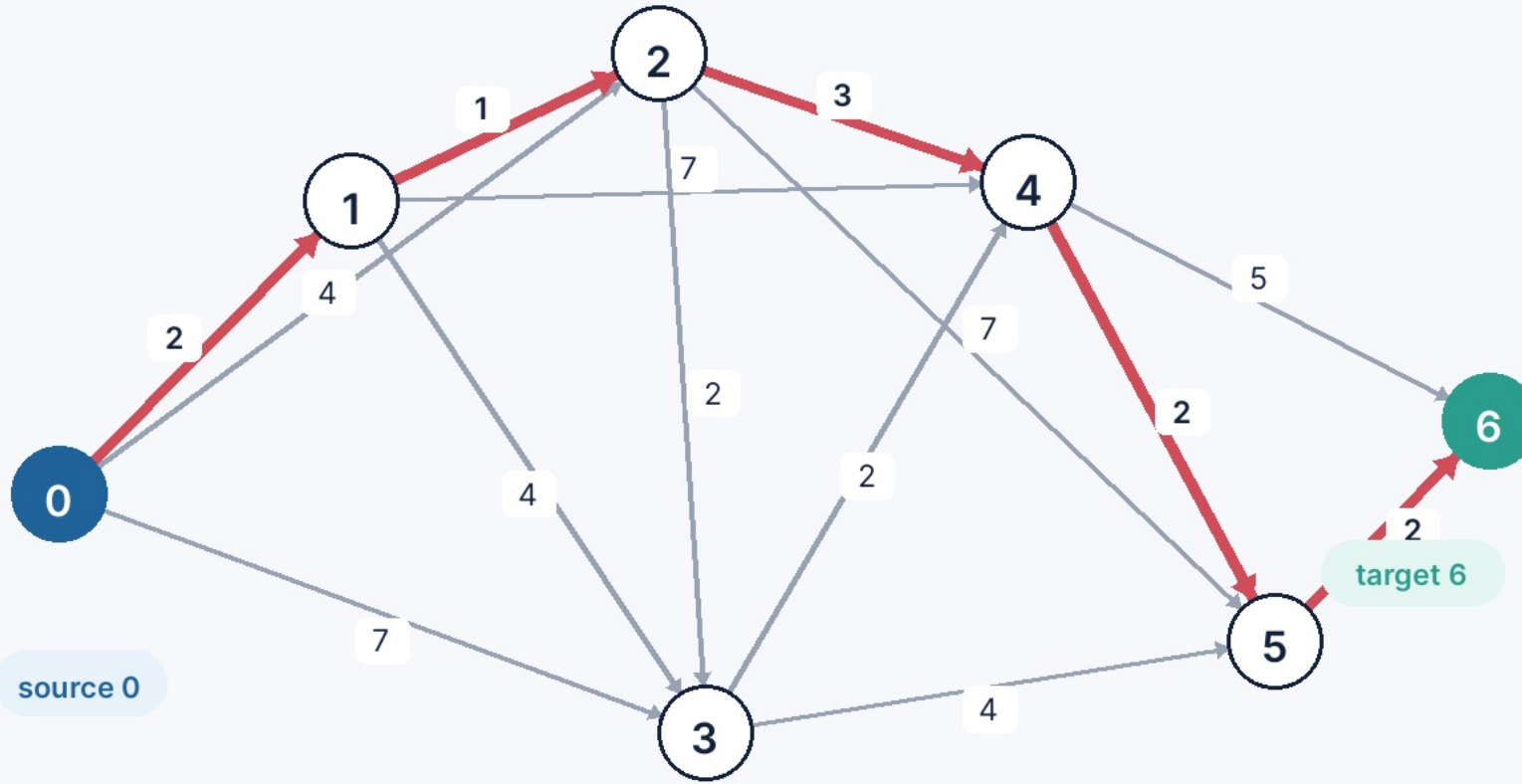
From Routing to QAOA: How Mixer Design and Circuit Depth Shape Feasibility and Optimality

Penalty-X vs Global-Grover QAOA on a weighted routing problem

RESEARCH QUESTION How do mixer design and QAOA depth affect the probability of finding feasible and optimal routes?



A small graph, a large binary space



DIRECTED GRAPH

7 nodes · 14 edges

source 0 → target 6

BINARY SEARCH SPACE

$2^{14} = 16,384$

one bit per directed edge

VALID DECODED ROUTES

20 bitstrings

only 0.122% of the space

UNIQUE OPTIMUM

$0 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 6$

routing cost = 10

A valid route sends exactly one unit of directed flow from source to target.

From edges to a routing QUBO

one directed edge $\rightarrow$ one bit

$x_e = 1$ selected
 $x_e = 0$ not selected

$$C(x) = \sum_e w_e x_e$$

DIRECTED FLOW BALANCE

$$\sum_{e \in \delta^+(v)} x_e - \sum_{e \in \delta^-(v)} x_e = b_v$$

$b_v = +1$ source
 $b_v = 0$ intermediate
 $b_v = -1$ target

SQUARE THE RESIDUAL

$$P_{\text{flow}}(x) = \sum_v \left(\sum_{\text{out}(v)} x_e - \sum_{\text{in}(v)} x_e - b_v \right)^2$$

$$Q_A(x) = \sum_e w_e x_e + A P_{\text{flow}}(x)$$

Frozen study: $A = 6$

Objective tells QAOA what is cheap.

Penalty tells QAOA what is valid.

Example: selecting only $0 \rightarrow 1$ leaves unmatched flow at node 1 and at the target, so the squared penalty is nonzero.

QUBO → Ising → QAOA

EXPANDED QUBO

$$Q(x) = c + \sum_i a_i x_i + \sum_{i < j} b_{ij} x_i x_j$$

Equivalent to $x^T Q x + \text{offset}$ after coefficient collection

BINARY → PAULI

$$x_i = \frac{1 - Z_i}{2}$$

$$|0\rangle \leftrightarrow x_i = 0 \quad |1\rangle \leftrightarrow x_i = 1$$

ISING COST HAMILTONIAN

$$H_C = c_0 I + \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j$$

Same energy landscape, now in quantum operators

$$U_C(\gamma) = e^{-i\gamma H_C}$$

Cost phase tags basis states by energy

$$|\psi_p\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell H_M} e^{-i\gamma_\ell H_C} |\psi_0\rangle$$

Graph

QUBO

Ising

Cost layer

Mixer

Measure

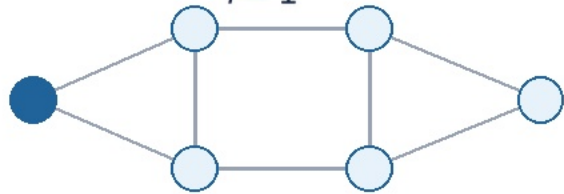
Route

The cost layer changes phases; the mixer turns phase differences into interference and probability redistribution.

Same cost landscape, two mixer geometries

PENALTY-X

$$H_M^X = \sum_{i=1}^{14} X_i, \quad U_M^X(\beta) = e^{-i\beta H_M^X}$$

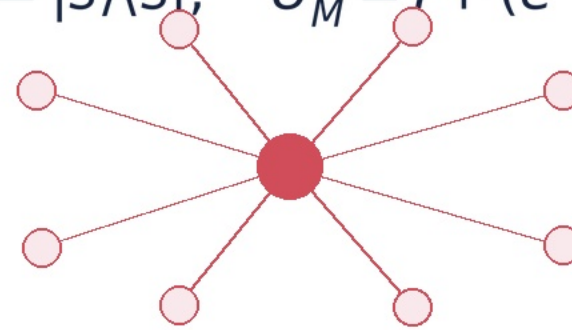


Single-bit flips
connect nearby
bitstrings

- Local, independent qubit rotations
- Explores the full 14-bit hypercube
- Feasibility comes only from the penalty

GLOBAL-GROVER

$$H_M^G = |s\rangle\langle s|, \quad U_M^G = I + (e^{-i\beta} - 1)|s\rangle\langle s|$$



One rank-one
update couples
the whole state

- Uniform $|s\rangle$ over all 16,384 states
- Global amplitude redistribution
- Full-space and not feasibility-preserving

Not standard Grover Search: there is no oracle that directly marks the optimal route; H_C supplies continuous cost phases.

The hybrid loop: COBYLA and CVaR

COBYLA = Constrained Optimization BY Linear Approximations

Initial angles

QAOA simulation

Objective value

COBYLA update

Derivative-free; it optimizes $\gamma_1 \dots \gamma_p$ and $\beta_1 \dots \beta_p$ — not the route directly.

VARIATIONAL DIMENSION

2p parameters

p=110 → 220 angles

FROZEN DEPTH BUDGET

$\max(120, 4p + 64)$

p=110 → 504 objective evaluations

MAIN-SWEEP OBJECTIVE

Expectation $E[H_C]$

same objective for both mixers

$$\langle H_C \rangle = \sum_x P(x) E(x) \quad \text{vs} \quad \text{CVaR}_\alpha = \text{mean energy in the lowest-energy } \alpha \text{ probability mass}$$

CVaR changes the classical objective, not the mixer. Separate p=110 study: Penalty-X 7/10 paired wins (mixed); Global-Grover 10/10.

What we measure — and how

p_{feas}

P(valid route)

How much mass reaches the feasible region?

p_{opt}

P(optimal route)

What is the direct single-shot success probability?

$p_{\text{opt}} \mid \text{feas}$

$p_{\text{opt}} / p_{\text{feas}}$

If feasible, how often is it the optimum?

$$p_{\text{opt}} = p_{\text{feas}} \times p_{\text{opt} \mid \text{feas}}$$

FROZEN MAIN SETUP

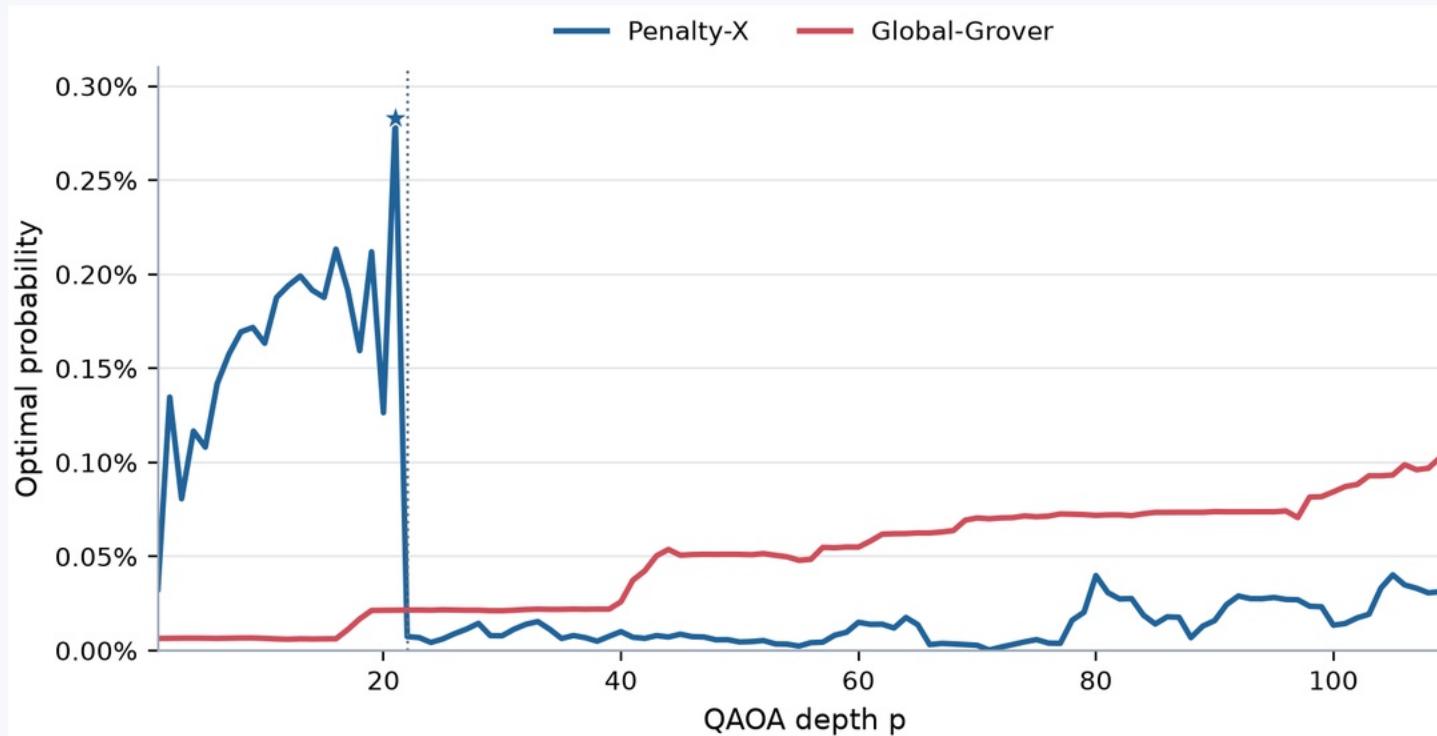
- 7 nodes · 14 edge qubits · $A=6$
- Penalty-X and Global-Grover
- depth $p = 1, \dots, 110$
- one optimization per method-depth
- deterministic layerwise continuation
- same full space, cost, decoder, and depth budget
- ideal NumPy complex128 statevectors

TOTAL

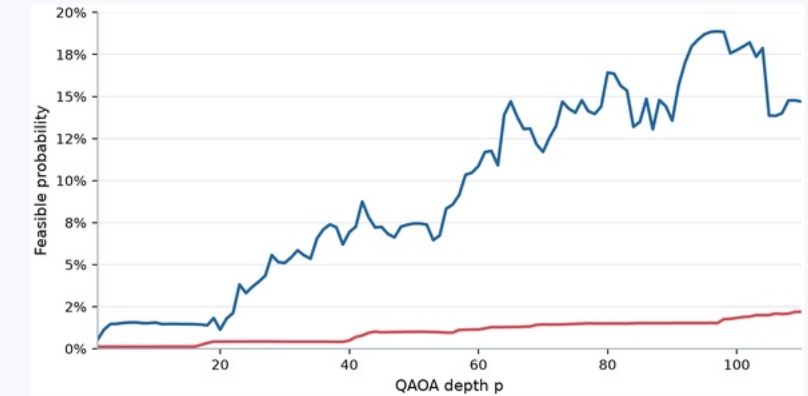
$$2 \times 110 = 220$$

Depth 1–110: where does probability go?

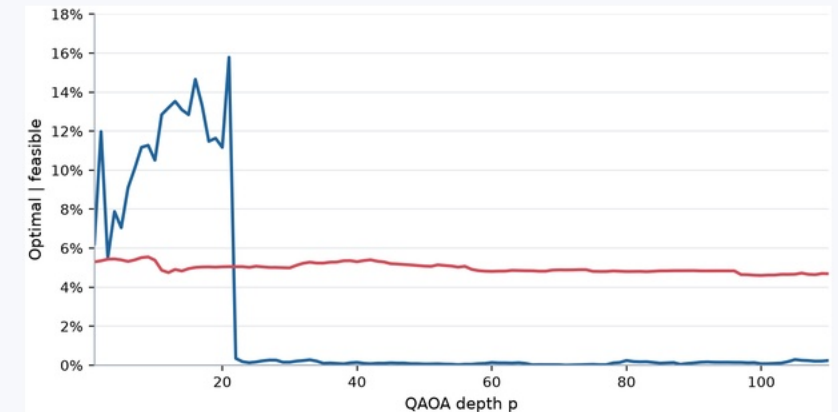
TOTAL OPTIMAL PROBABILITY



FEASIBLE PROBABILITY



OPTIMAL GIVEN FEASIBLE



Penalty-X peak: $p=21$, 0.2833% · first crossover: $p=22$ · Global-Grover peak: $p=110$, 0.1030%

Conditional quality is not total success

$$p_{\text{opt}} = p_{\text{feas}} \times p_{\text{opt} \mid \text{feas}}$$

PENALTY-X · p=110

P_FEAS

14.6970%

×

P_OPT | FEAS

0.2435%

=

P_OPT

0.03579%

same-depth total success

GLOBAL-GROVER · p=110

P_FEAS

2.1946%

×

P_OPT | FEAS

4.6923%

=

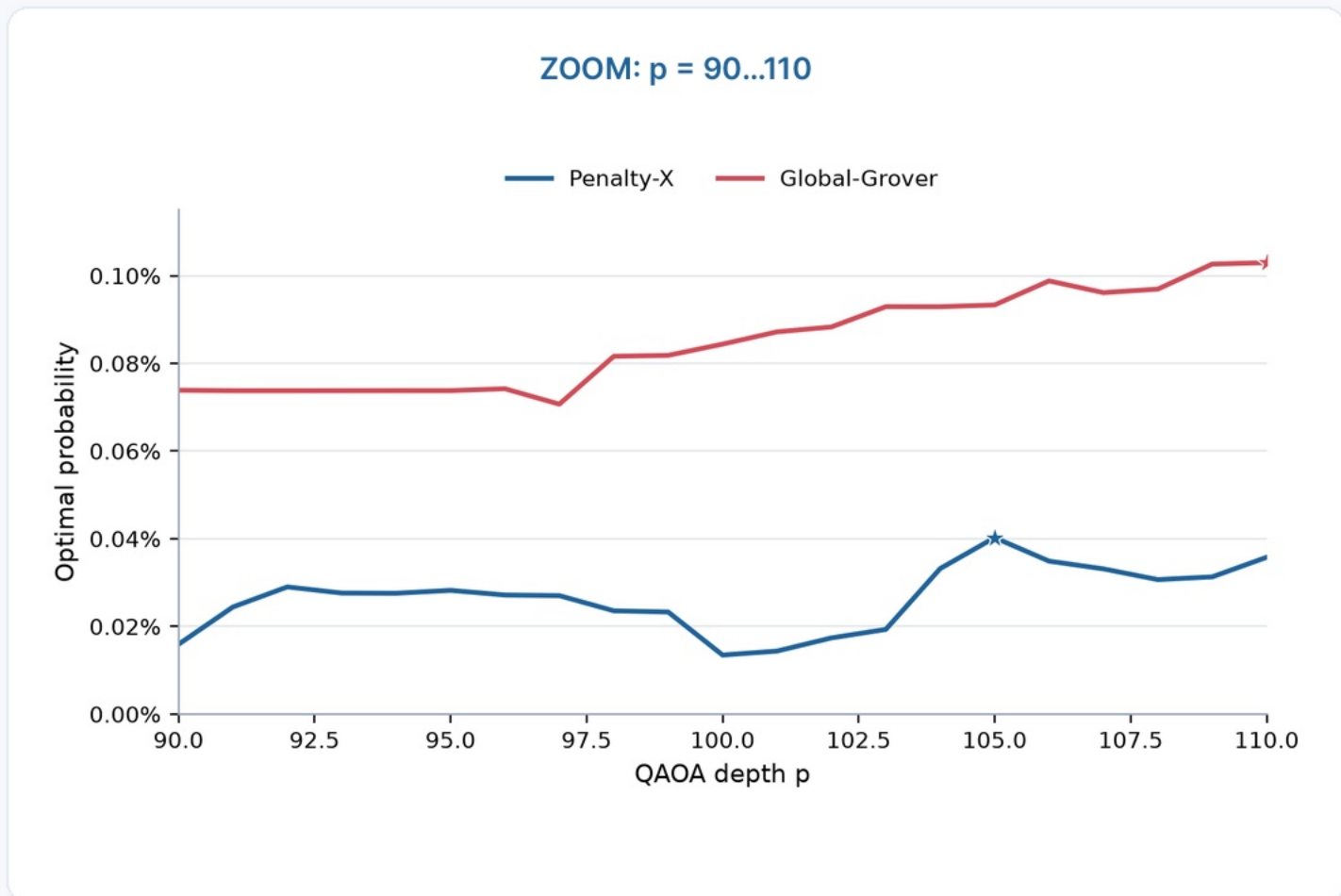
P_OPT

0.10298%

2.88× Penalty-X at p=110

Global-Grover is 19.3× stronger conditionally, but has 6.7× less feasible mass. It wins at p=110, yet its best never exceeds Penalty-X's earlier 0.2833% peak.

Deeper is not automatically better



OBSERVED NEAR $P \approx 100$

No sharp transition

trajectories remain non-monotonic

$P=90 \dots 110$ DIRECTION CHANGES

6 vs 9

Penalty-X vs Global-Grover

AT $P=110$

220 parameters

504 COBYLA evaluations

OPTIMIZATION STATUS

All 220 cells

budget-limited; $p_{\text{opt}} < 10\%$

Within this instance and finite budget, depth to 110 reveals no sharp Grover-scale success transition; ansatz and optimizer difficulty cannot be separated.

Scientific computing and validation



REPRODUCIBLE COMPUTING STACK

- Python + NumPy exact statevectors
- SciPy COBYLA parameter optimization
- NetworkX exact path reference
- Explicit QUBO construction and Ising mapping
- CSV / JSON checkpoints and canonical tables
- Matplotlib figures + CLI scripts + Git

CURRENT VERIFICATION

- 167 tests passing
- 23 visualization checks passing
- 43 / 43 depth-artifact checks
- 43 / 43 CVaR-artifact checks
- max QUBO-Ising error = 0
- route 0→1→2→4→5→6, cost 10

Smoke checks: `src/main.py` · `Penalty-X` command · `course-final` command · read-only visualization replay

Takeaways and limitations

FIVE TAKEAWAYS

- 1 Complete pipeline: graph $\rightarrow$ QUBO $\rightarrow$ Ising $\rightarrow$ QAOA $\rightarrow$ decoded routes.
- 2 Penalty-X and Global-Grover create different probability distributions.
- 3 Penalty-X puts substantially more mass on feasible routes.
- 4 Global-Grover has stronger conditional optimum concentration at 89/110 depths.
- 5 **Conditional concentration alone does not determine total success.**

LIMITATIONS

- one fixed 7-node instance
- ideal simulator; no QPU evidence
- finite, exhausted optimizer budgets
- high-depth optimization is difficult
- experiment-specific Global-Grover
- CVaR limited to tested protocol
- no quantum-advantage claim

$$p_{\text{opt}} = p_{\text{feas}} p_{\text{opt} \mid \text{feas}}$$

Feasibility, optimal concentration, and optimization difficulty are separate effects — a mixer can improve one without improving the others.

Backup A — Why subtract b_v ?

$$\text{out}(v) - \text{in}(v) = b_v \implies \text{out}(v) - \text{in}(v) - b_v = 0$$

SOURCE

$b = +1$

one more outgoing
than incoming

INTERMEDIATE

$b = 0$

incoming equals
outgoing

TARGET

$b = -1$

one more incoming
than outgoing

VALID: $0 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 6$

source: $+1 - (+1) = 0$

node 2: $(1 - 1) - 0 = 0$

target: $-1 - (-1) = 0$

INVALID: select only $0 \rightarrow 1$

source residual = 0

node 1 residual = -1

target residual = +1 $\rightarrow$ penalty 2

Squaring makes every nonzero flow residual contribute a positive energy penalty.

Backup B — QUBO matrix and Ising mapping

TWO EQUIVALENT QUBO VIEWS

$$c + \sum_i a_i x_i + \sum_{i < j} b_{ij} x_i x_j$$

$$x^T Q x + \text{offset}$$

SUBSTITUTE BINARY OPERATORS

$$x_i = \frac{I - Z_i}{2} \quad x_i x_j = \frac{1}{4}(I - Z_i - Z_j + Z_i Z_j)$$

Identity, single-Z, and ZZ terms appear after expansion.

$$H_C = c_0 I + \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j$$

QUBO CONSTANT

12

for A=6

NONZERO PAIRS

46

same in QUBO and Ising

EXHAUSTIVE ENERGY ERROR

0 across 16,384 states

exact Fraction arithmetic

Backup C — What is COBYLA?

Constrained Optimization BY Linear Approximations

- classical, derivative-free optimizer
- proposes QAOA angle vectors
- uses objective evaluations, not gradients
- does not choose routes directly
- budget limits are not convergence proofs

WHY DEPTH IS HARD

$$n_{\theta} = 2p$$

$p = 110 \rightarrow 220$ coupled angles

budget = $\max(120, 4p + 64)$

$p = 110 \rightarrow 504$ evaluations

All 220 depth-sweep cells reached their budget.

A deeper ansatz can be more expressive while simultaneously being harder for the classical optimizer to tune.

Backup D — Global-Grover is not standard Grover Search

STANDARD GROVER SEARCH

THIS GLOBAL-GROVER QAOA

Goal Find a marked state

Optimize a cost distribution

Problem phase Binary oracle marks target

Continuous $e^{-i \sum c}$ cost phase

Mixing step Fixed diffusion

Variational rank-one mixer

Angles Prescribed iterations

γ and β optimized by COBYLA

Optimal route known? Oracle supplies marking

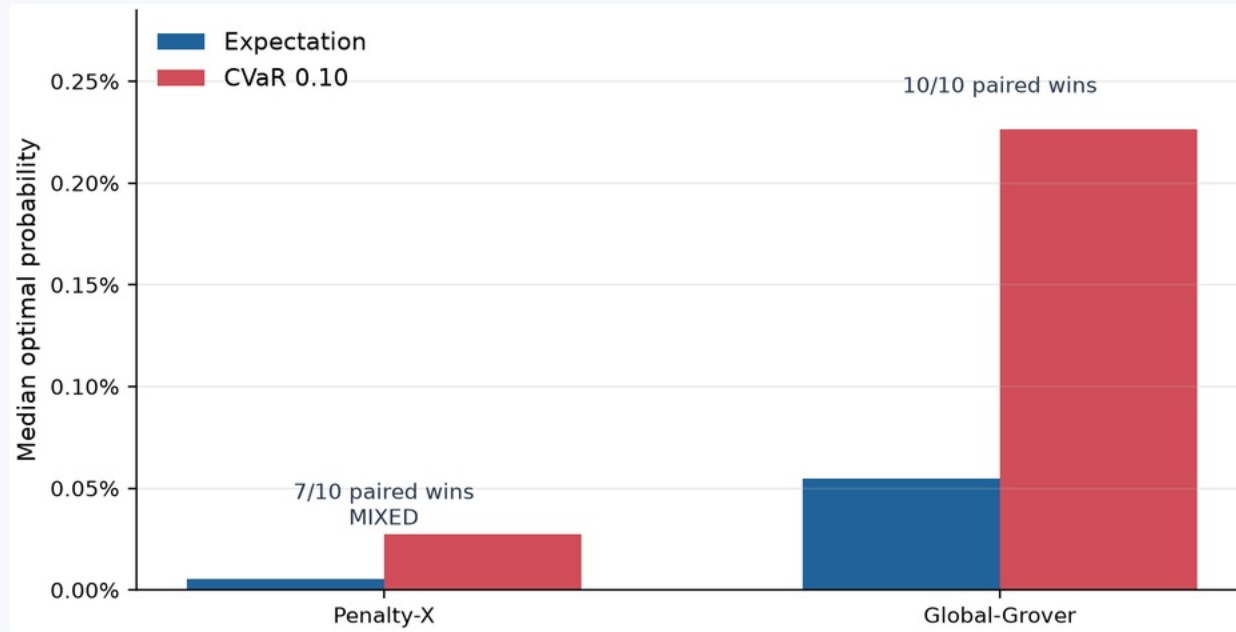
No optimum-marking oracle

Feasibility Depends on oracle/domain

Full 16,384-state space; not preserved

The familiar $\pi/4 \cdot \sqrt{N}$ iteration count is exploratory context only; it is not a prediction for this variational continuous-cost circuit.

Backup E — CVaR robustness study



PENALTY-X · $p=110$

7/10 wins · median ratio 2.971 · MIXED

GLOBAL-GROVER · $p=110$

10/10 wins · median ratio 3.929

FROZEN PROTOCOL

α in {0.05, 0.10, 0.25, 0.50}
plus expectation ($\alpha=1$)
 $p = 50, 100, 110 \cdot 140$ cells

Supported mechanism: feasible-region amplification, not consistently stronger optimal concentration within the feasible set. All cells were budget-exhausted; absolute p_{opt} remained small.